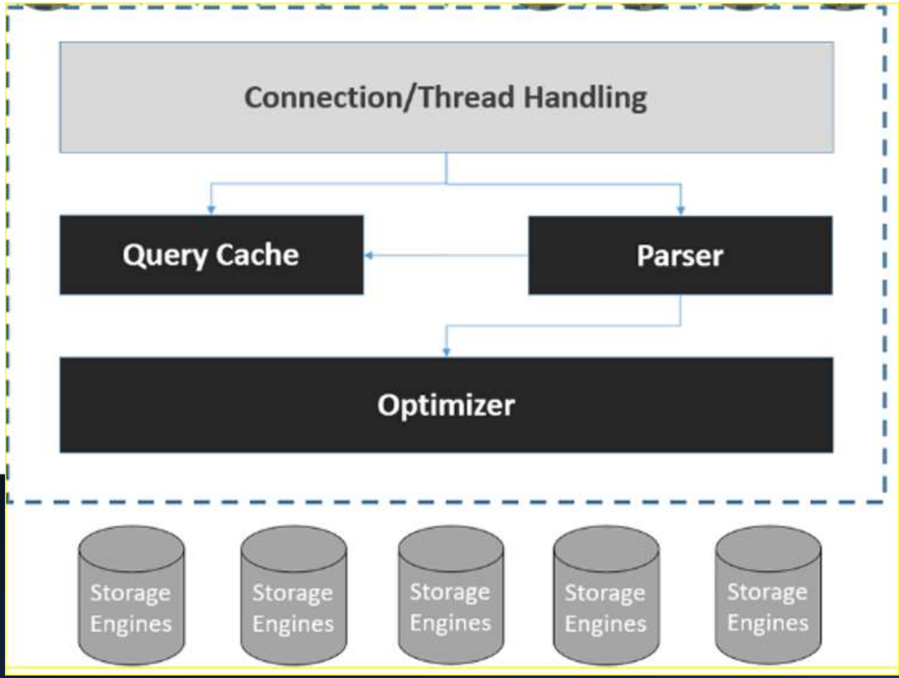
Basic MySQL Architecture and key components.



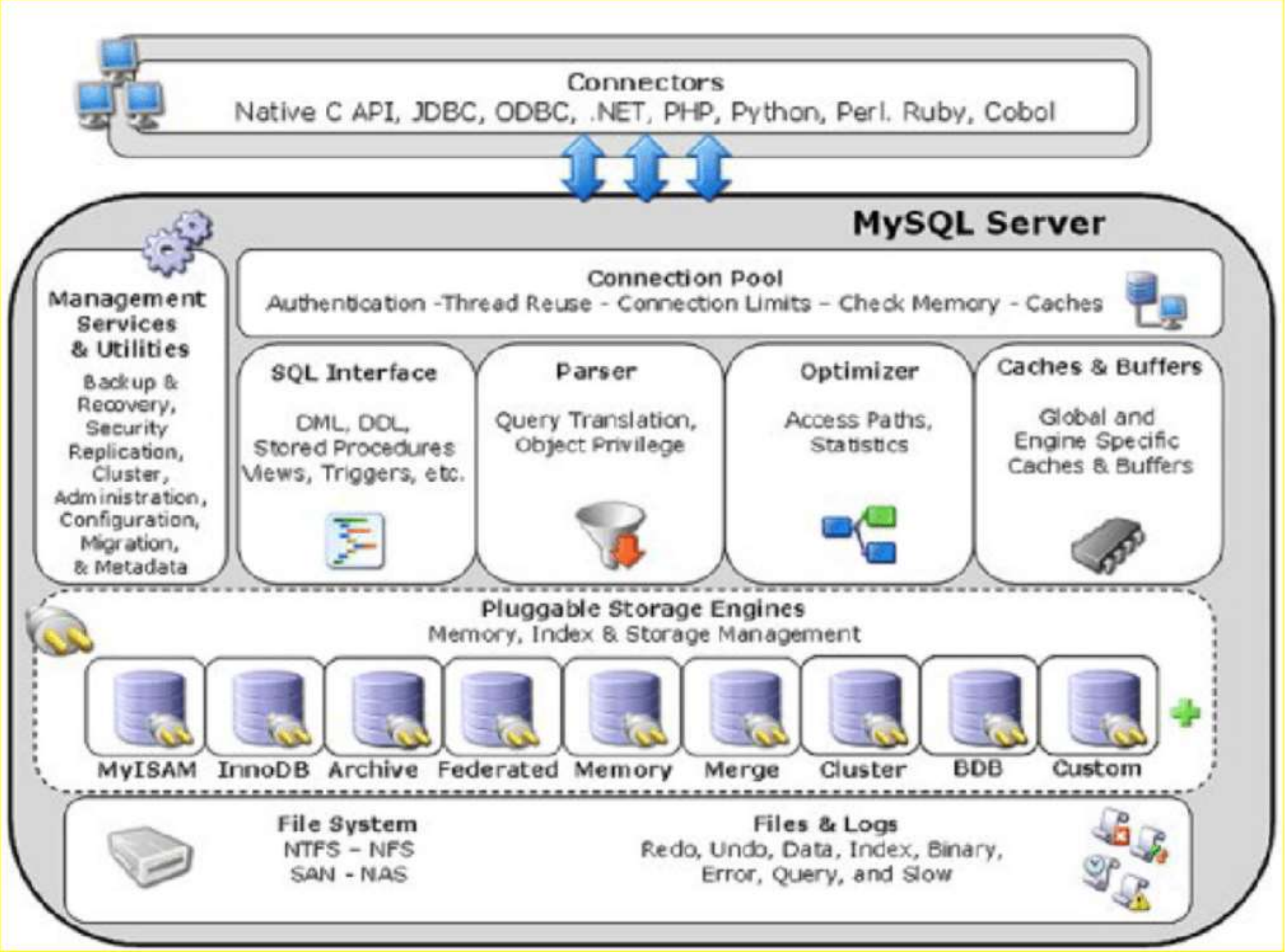
The MySQL architecture describes how the different components of a MySQL system relate to one another. The MySQL architecture is basically a client – server system. MySQL database server is the server and the applications which are connecting to MySQL database server are clients.

- **Application Layer**
  - User Interface
  - Client connect to server
- **Logical Layer**
  - Query processing
  - Transactions Management
  - Recovery Management
  - Storage Management
- **Physical Layer**
  - Storage Engine
  - InnoDB
  - MyISAM
  - CSV
  - NDB
  - Archive
  - Memory
  - Merge
  - Customer

File System Supported by MySQL - XFS, NTFS, Ext4, NFS, SAN, NFS

Files and Logs - Redu, Undo, Data files, Index, Binary logs, error logs, slow query logs

Advance MySQL Architecture



Types of MySQL Logs

MySQL Server has several **logs** that can help you find out what activity is taking place

- MySQL Error logs
- MySQL binary logs
- MySQL Slow query logs
- MySQL general logs
- MySQL Event Logs
- Redo and undo logs etc.

- Default Data Directories ➔ /var/lib/mysql

```
[root@mongodb1 ~]# ls -lrth /var/lib/mysql
```

- Default MySQL configurations files ➔ /etc/my.cnf

```
[root@mongodb1 ~]# ls -lrth /etc/my.cnf
```

- Default MySQL error log file

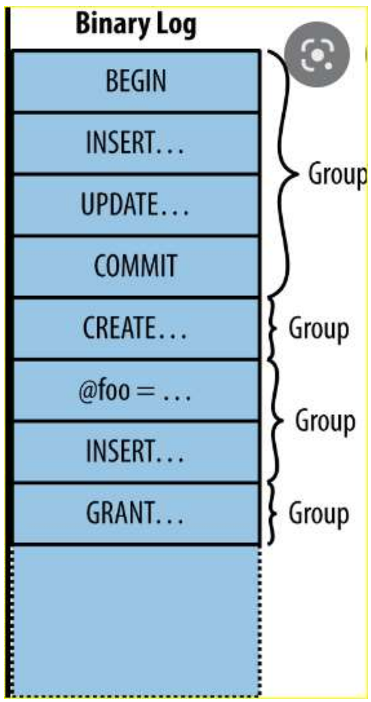
```
[root@mongodb1 ~]# ls -lrth /var/log/mysqld.log
```

- Default MySQL Port

```
[root@mongodb1 ~]# netstat -tulnp | grep -i mysql
tcp6       0      0 :::3306                 :::*                    LISTEN      1805/mysqld
```

What is MySQL binary logs?

The binary log is a set of log files that contain information about data modifications made to a MySQL server instance



The binary log is a set of log files that contain information about data modifications in a MySQL server instance. Whatever statement or Changes happening to the MySQL DB (Insert, Update, delete, drop, delete and etc) that will be getting stored in binary logs

The binary log has two important purposes:

1. For replication,
2. Data recovery operations

- **How to enable the binary log**

  --log-bin

  We need pass in config file or in command line we need to specify

- **In addition to above points the binary log also contains following information**

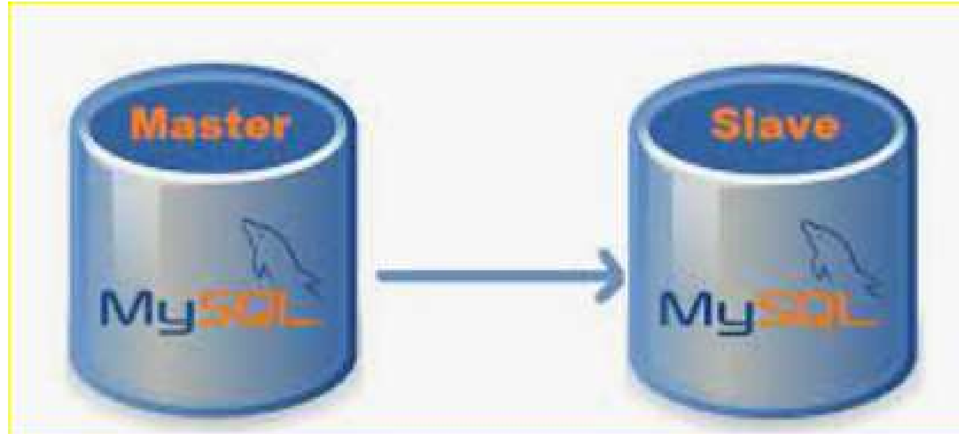
  1. Information about the state of the server that is needed to reproduce statements correctly

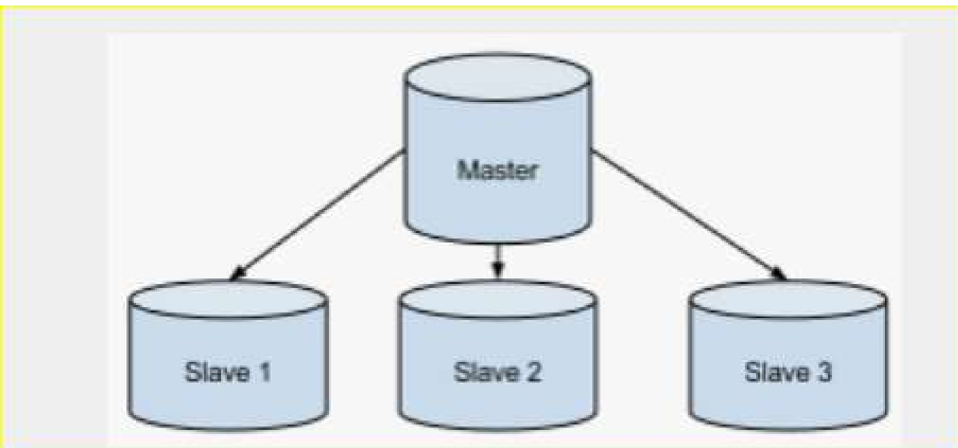
  2. Error codes

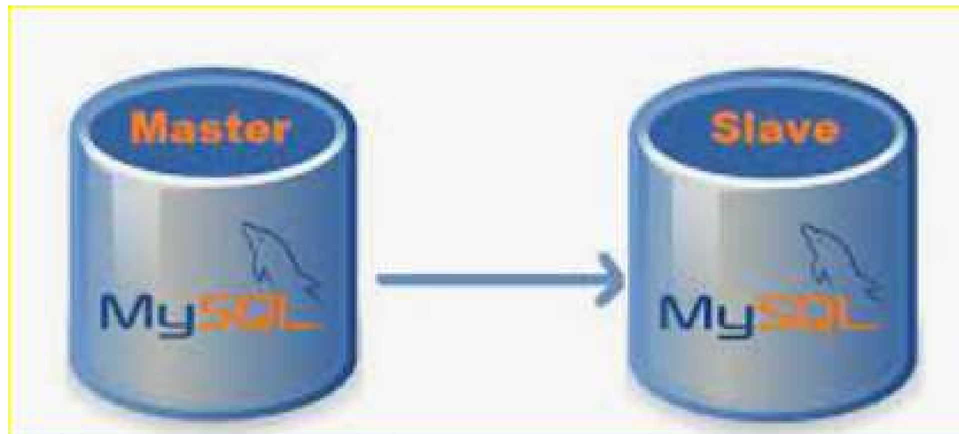
  3. Metadata needed for the maintenance of the binary log itself -

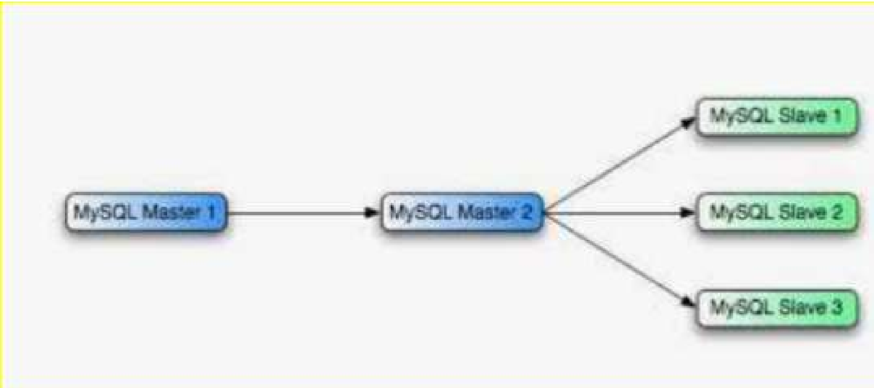
Types of MySQL instances?

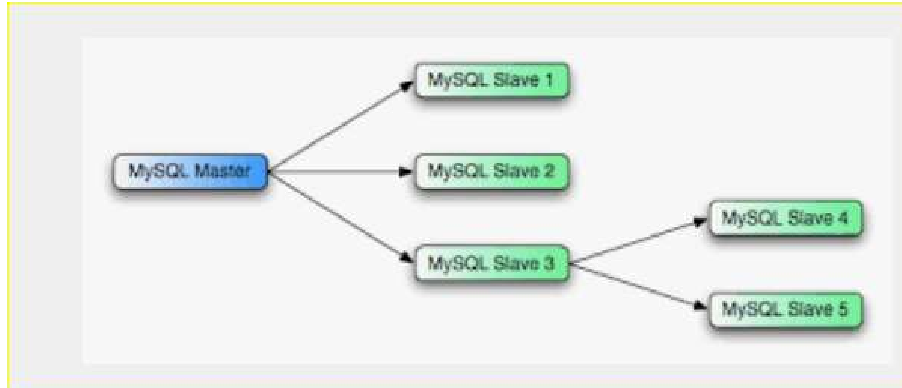
- Standalone
- Master- Slave
- Master- Master
- Chain Replication
- Multi Source replication

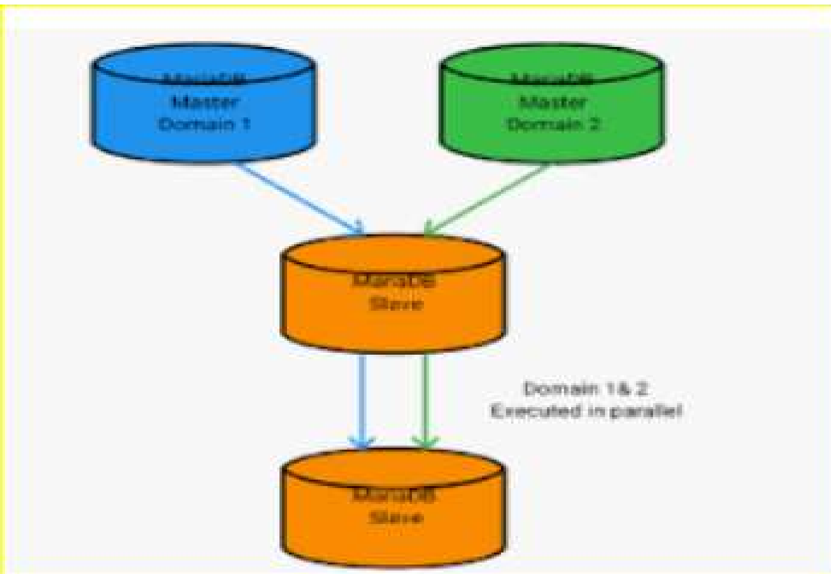


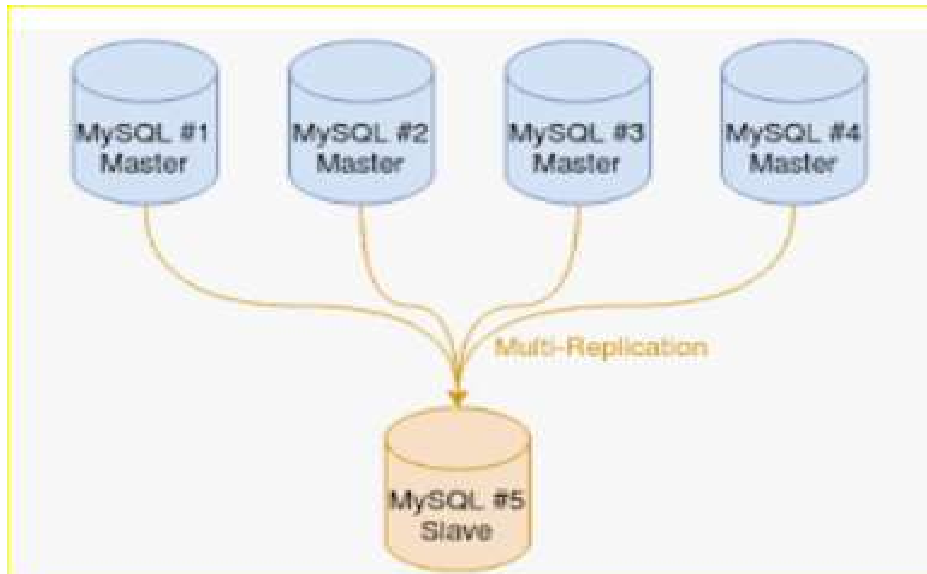





MySQL Master 1
MySQL Master 2
MySQL Slave 1
MySQL Slave 2
MySQL Slave 3


MySQL Master
MySQL Slave 1
MySQL Slave 2
MySQL Slave 3
MySQL Slave 4
MySQL Slave 5


Master
Domain 1
Master
Domain 2
MariaDB
Slave
Domain 1 & 2
Executed in parallel
MariaDB
Slave


MySQL #1
Master
MySQL #2
Master
MySQL #3
Master
MySQL #4
Master
Multi-Replication
MySQL #5
Slave

```
[root@mongodb1 ~]# ls -lrth /var/lib/mysql
total 121M
-rw-r----- 1 mysql mysql  48M Sep  1  2021 ib_logfile1
-rw-r----- 1 mysql mysql   56 Sep  1  2021 auto.cnf
-rw------- 1 mysql mysql 1.7K Sep  1  2021 ca-key.pem
-rw-r--r-- 1 mysql mysql 1.1K Sep  1  2021 ca.pem
-rw------- 1 mysql mysql 1.7K Sep  1  2021 server-key.pem
-rw-r--r-- 1 mysql mysql 1.1K Sep  1  2021 server-cert.pem
-rw------- 1 mysql mysql 1.7K Sep  1  2021 client-key.pem
-rw-r--r-- 1 mysql mysql 1.1K Sep  1  2021 client-cert.pem
-rw-r--r-- 1 mysql mysql  452 Sep  1  2021 public_key.pem
-rw------- 1 mysql mysql 1.7K Sep  1  2021 private_key.pem
drwxr-x--- 2 mysql mysql 8.0K Sep  1  2021 performance_schema
drwxr-x--- 2 mysql mysql 4.0K Sep  1  2021 mysql
drwxr-x--- 2 mysql mysql 8.0K Sep  1  2021 sys
-rw-r----- 1 mysql mysql  306 Jul  9 02:51 ib_buffer_pool
-rw------- 1 mysql mysql    5 Jul  9 02:51 mysql.sock.lock
srwxrwxrwx 1 mysql mysql    0 Jul  9 02:51 mysql.sock
-rw-r----- 1 mysql mysql  12M Jul  9 02:52 ibtmp1
-rw-r----- 1 mysql mysql  12M Jul  9 02:52 ibdata1
-rw-r----- 1 mysql mysql  48M Jul  9 02:52 ib_logfile0
```